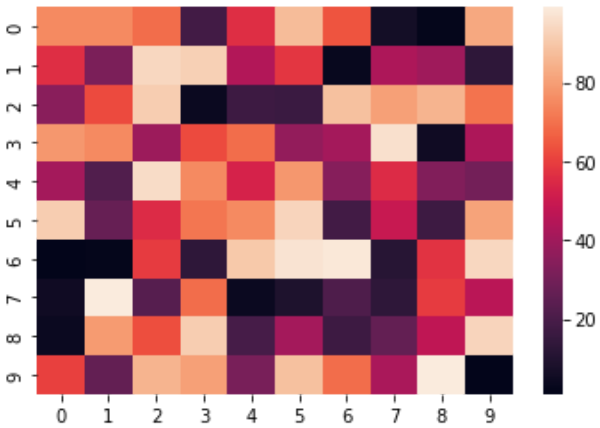
Heat map

sns.heatmap(data,vmin,vmax,cmap,annot,linewidth,linecolor,cbar,xticklabels,ytic

In [5]:

```
# importing the modules

import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# generating 2-D 10*10 matrix of random numbers from 1 to 100

data=np.random.randint(low=1,high=100,size=(10,10))
print("The data to be plotted:\n")
print(data)

# plotting heatmap

sns.heatmap(data)

# show ploted Heatmap
plt.show()
```

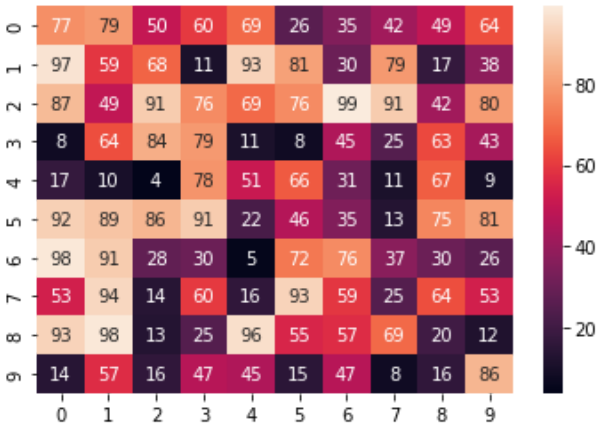
```
The data to be plotted:

[[75 75 69 18 56 87 64  6  2 82]
 [56 32 94 92 44 58  3 43 40 13]
 [35 62 91  4 17 16 88 80 85 70]
 [78 75 39 62 69 37 41 96  5 43]
 [41 22 95 75 53 78 34 55 33 30]
 [91 27 55 71 75 93 18 49 17 81]
 [ 1  2 59 13 90 97 98 11 57 94]
 [ 5 99 23 69  4  9 21 13 59 46]
 [ 4 79 63 91 19 41 17 26 47 93]
 [60 26 85 80 31 88 69 42 99  1]]
```



In [4]:

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# generating 2-D 10*10 matrix of random numbers from 1 to 100

data=np.random.randint(low=1,high=100,size=(10,10))
print("The data to be plotted:\n")
print(data)

# plotting heatmap

sns.heatmap(data,annot=True)

# show ploted Heatmap
plt.show()
```

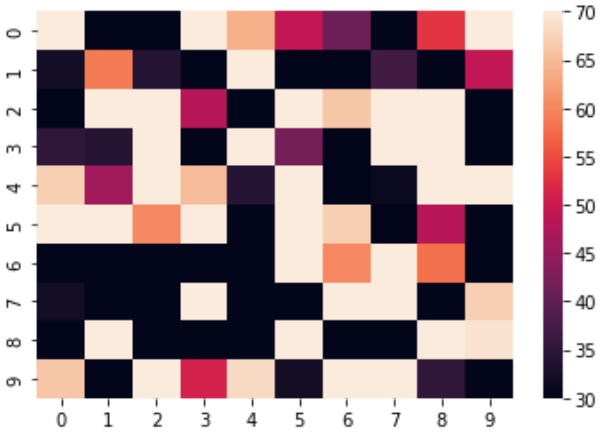
```
The data to be plotted:

[[77 79 50 60 69 26 35 42 49 64]
 [97 59 68 11 93 81 30 79 17 38]
 [87 49 91 76 69 76 99 91 42 80]
 [ 8 64 84 79 11  8 45 25 63 43]
 [17 10  4 78 51 66 31 11 67  9]
 [92 89 86 91 22 46 35 13 75 81]
 [98 91 28 30  5 72 76 37 30 26]
 [53 94 14 60 16 93 59 25 64 53]
 [93 98 13 25 96 55 57 69 20 12]
 [14 57 16 47 45 15 47  8 16 86]]
```



In [6]:

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# generating 2-D 10*10 matrix of random numbers from 1 to 100

data=np.random.randint(low=1,high=100,size=(10,10))
print("The data to be plotted:\n")
print(data)

# plotting heatmap

sns.heatmap(data,vmin=30,vmax=70)

# show ploted Heatmap
plt.show()
```

```
The data to be plotted:

[[73 27 12 72 64 49 41 20 53 77]
 [32 59 34 24 74 12 14 37 29 49]
 [ 5 71 71 48 17 75 66 73 77 14]
 [35 34 93 29 75 42  3 86 83  5]
 [67 46 70 65 34 78 11 31 85 74]
 [99 79 60 71  4 74 67 17 48 22]
 [ 1 14  2 15 25 93 60 86 58  3]
 [32 15 23 88 26 17 82 79 11 67]
 [ 5 82 30 23 12 82 17 16 93 69]
 [66 17 87 51 68 32 94 74 35 26]]
```



In [7]:

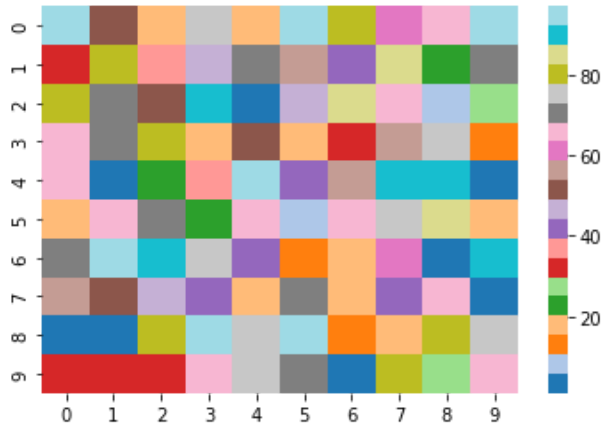
```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# generating 2-D 10*10 matrix of random numbers from 1 to 100

data=np.random.randint(low=1,high=100,size=(10,10))
print("The data to be plotted:\n")
print(data)

# plotting heatmap

sns.heatmap(data,vmin=30,vmax=70,annot=True)

# show ploted Heatmap
plt.show()
```

```
The data to be plotted:

[[76 48 45 42 68 84  9 35 95 37]
 [ 1 10 63 73 96 29 31 58 21 17]
 [ 8 28  7 45 17 72 46  3 84 94]
 [13  6 13  4 63  6 19 15 18 42]
 [78 10 30  3 55 16 73 85 73 92]
 [72 74 83  3 40 90 99 81 14 31]
 [53 23 80 31 56 62 62 84 49 44]
 [32 29 10 50 84 22 75 68 92 15]
 [74 51 39 62  3  8 14 12  9 17]
 [25 38 62  5 79 49 74 16 61 54]]
```

In [8]:

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# generating 2-D 10*10 matrix of random numbers from 1 to 100

data=np.random.randint(low=1,high=100,size=(10,10))
print("The data to be plotted:\n")
print(data)

# plotting heatmap

sns.heatmap(data,cmap="tab20")

# show ploted Heatmap
plt.show()
```

```
The data to be plotted:

[[97 50 18 73 16 94 82 62 66 94]
 [31 78 37 45 70 57 40 84 21 72]
 [79 72 49 91  1 47 87 67 10 26]
 [67 72 80 16 52 17 31 54 77 14]
 [65  2 21 39 94 41 56 90 88  2]
 [17 67 69 22 65  6 67 75 84 18]
 [72 95 90 74 41 14 19 63  5 92]
 [58 49 46 41 16 69 17 43 67  1]
 [ 1  5 79 96 75 94 14 16 78 73]
 [32 31 30 67 77 70  1 82 29 68]]
```



In [32]:

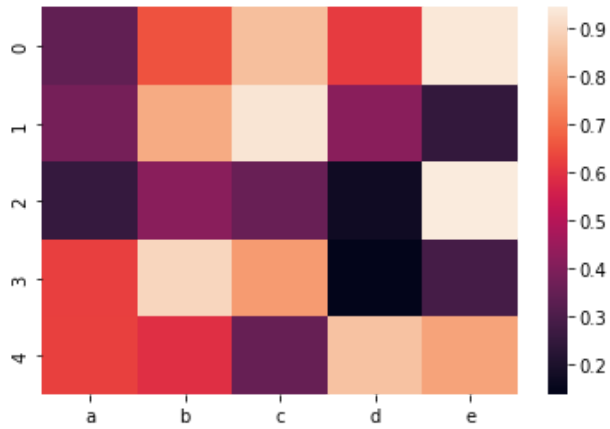
```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# generating 2-D 10*10 matrix of random numbers from 1 to 100

data=np.random.randint(low=1,high=100,size=(10,10))
print("The data to be plotted:\n")
print(data)

# plotting heatmap

sns.heatmap(
    data,
    vmin=10,
    vmax=70,
    cmap="Pastel1",
    annot=True,
    linewidth=1,
    linecolor="black",
    cbar=False,
    xticklabels=False,
    yticklabels=False
)


# show ploted Heatmap
plt.show()
```

```
The data to be plotted:

[[10 25 96  6 22 13 20 37 89 15]
 [27  3 50 52 13 29  9 14 59 78]
 [87 61 95 18 62 41 87 82 62 34]
 [44 58 12 57 96 75 61 77  7  9]
 [66 70 21 28 57 35 68 97 92 83]
 [13 26 34 68 62 70 68 63 35 84]
 [ 3 19  8 50 27 39 70 28 87 40]
 [25 94 21 90  8 25 18 58 57 86]
 [97 78  6 14 17 24  1 73 45 35]
 [ 5 63 10 64 64 58 11 67 43 63]]
```

10	25	96	6	22	13	20	37	89	15
27	3	50	52	13	29	9	14	59	78
87	61	95	18	62	41	87	82	62	34
44	58	12	57	96	75	61	77	7	9
66	70	21	28	57	35	68	97	92	83
13	26	34	68	62	70	68	63	35	84
3	19	8	50	27	39	70	28	87	40
25	94	21	90	8	25	18	58	57	86
97	78	6	14	17	24	1	73	45	35
5	63	10	64	64	58	11	67	43	63

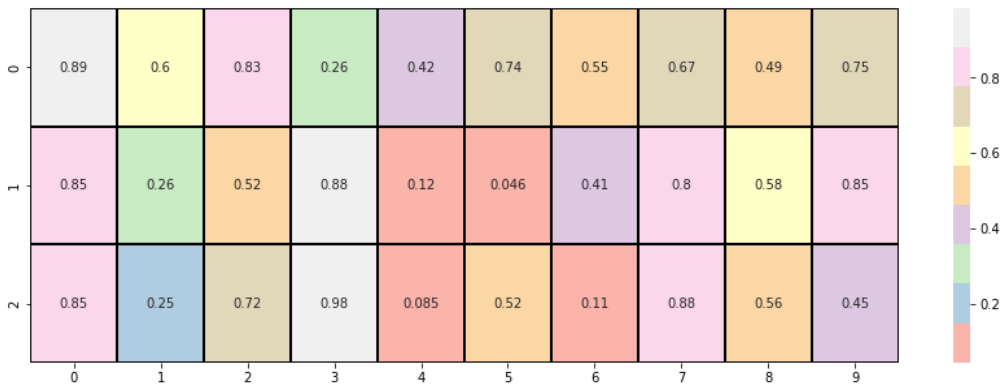
In [11]:
```
import numpy as np
import seaborn as sns
import pandas as pd

df=pd.DataFrame(np.random.random((5,5)),columns=["a","b","c","d",'e'])
print(df)
p1=sns.heatmap(df)
```

```
          a         b         c         d         e
0  0.341176  0.652949  0.850282  0.616401  0.935868
1  0.376990  0.811468  0.931960  0.414642  0.245261
2  0.253553  0.415481  0.351958  0.169259  0.944695
3  0.620564  0.898148  0.778093  0.136824  0.283995
4  0.623962  0.592687  0.349442  0.857008  0.795054
```



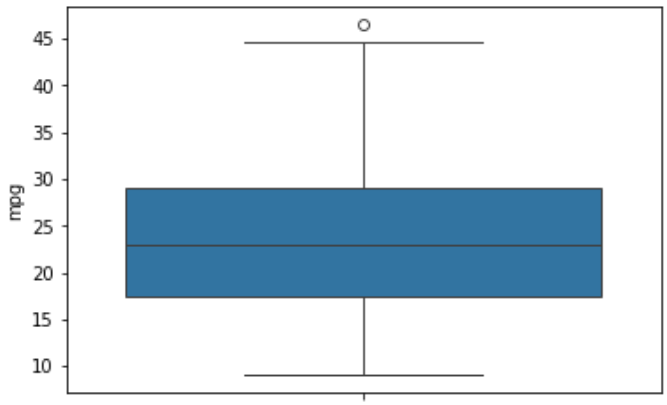
In [18]:
```
import numpy as np
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

plt.figure(figsize=(15,5))
df=pd.DataFrame(np.random.random((3,10)))
print(df)
p1=sns.heatmap(df,cmap="Pastel1",annot=True,linewidth=2,linecolor="black",cbar=True,x
ticklabels=True,yticklabels=True)
```

```
          0         1         2         3         4         5         6  \
0  0.893402  0.598283  0.828284  0.256656  0.416291  0.740202  0.553545
1  0.846169  0.258722  0.517467  0.882518  0.119383  0.045796  0.406350
2  0.848780  0.249734  0.718913  0.984252  0.085159  0.515854  0.109184

          7         8         9
0  0.674017  0.490265  0.746767
1  0.803769  0.581923  0.853554
2  0.877194  0.556358  0.449952
```

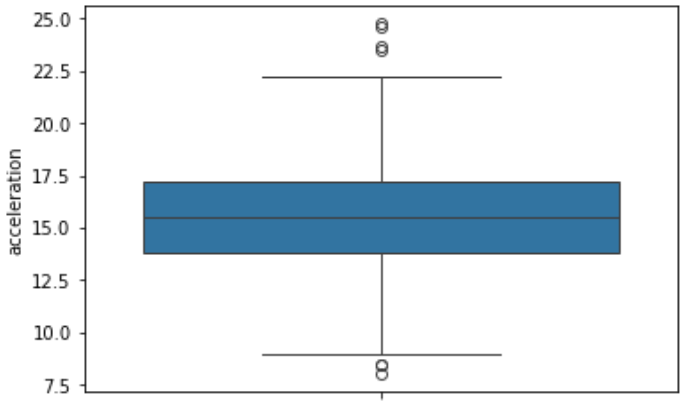


Box Plot

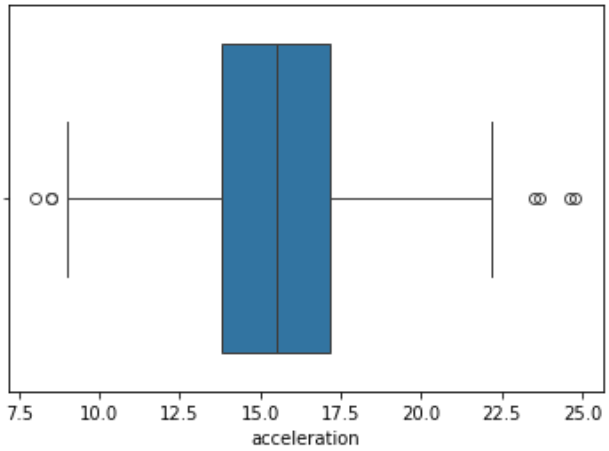
In [20]:
```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df=pd.read_csv("auto-mpg.csv")
sns.boxplot(y=df["mpg"])
plt.show()
```



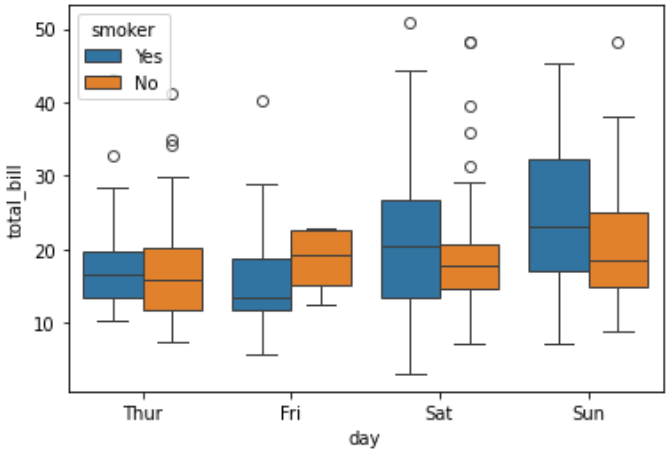
In [21]:
```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df=pd.read_csv("auto-mpg.csv")
sns.boxplot(y=df["acceleration"])
plt.show()
```



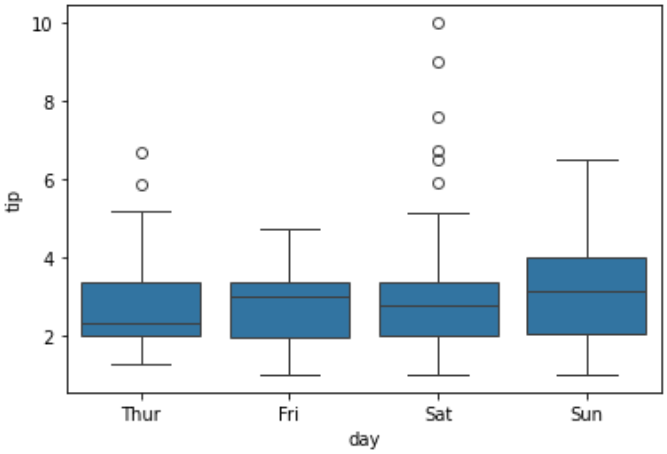
In [22]:
```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df=pd.read_csv("auto-mpg.csv")
sns.boxplot(x=df["acceleration"])
plt.show()
```



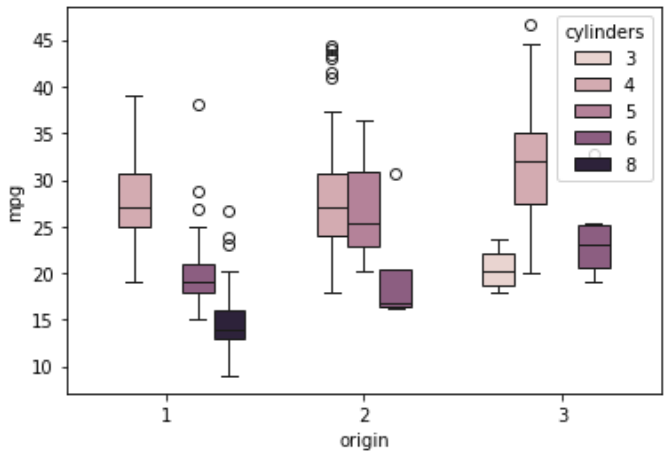
In [26]:
```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df=sns.load_dataset("tips")
# df=pd.read_csv("tips.csv")
sns.boxplot(x="day",y="total_bill",hue="smoker",data=df)
plt.show()
```



In [27]:
```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df=sns.load_dataset("tips")
# df=pd.read_csv("tips.csv")
sns.boxplot(x="day",y="tip",data=df)
plt.show()
```



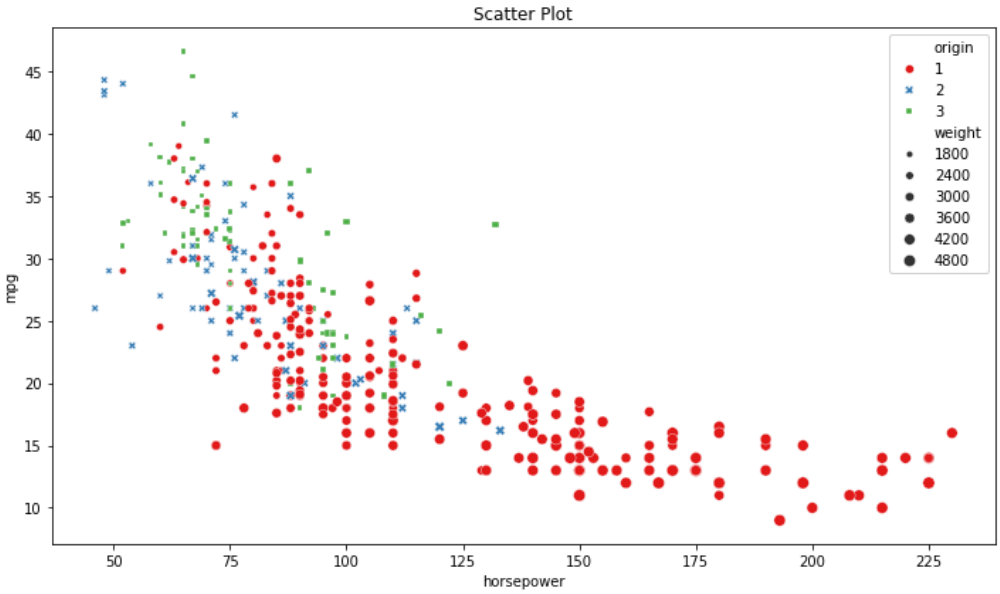
In [33]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df=pd.read_csv("auto-mpg.csv")
sns.boxplot(x="origin",y="mpg",hue="cylinders",data=df)
plt.show()
```



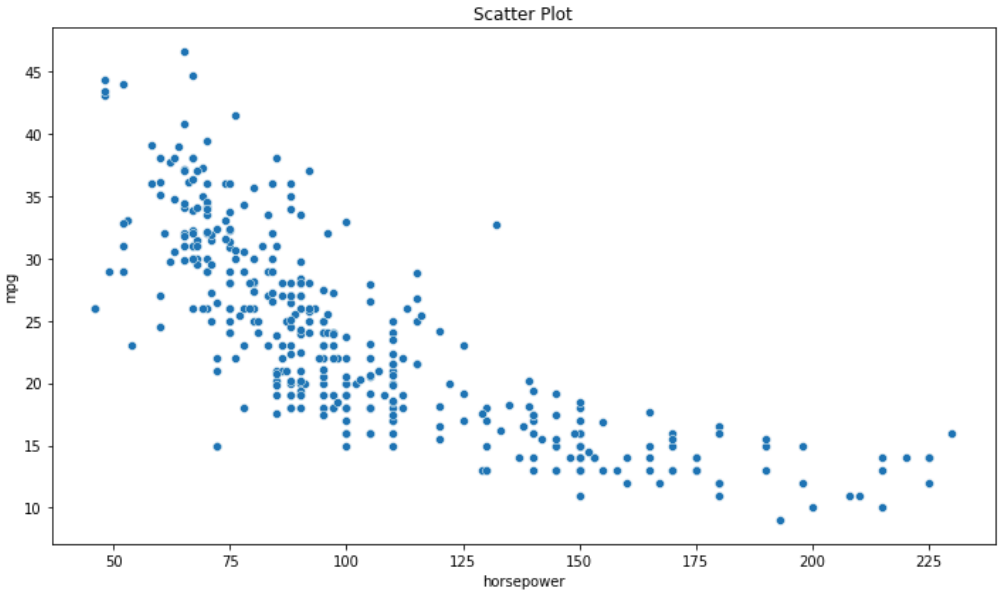
Scatter plot

In [34]:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sd=pd.read_csv("auto-mpg.csv")
df["horsepower"]=pd.to_numeric(df["horsepower"],errors="coerce")
df=df.dropna(subset=["horsepower"])
plt.figure(figsize=(10,6))

sns.scatterplot(
    data=df,
    x="horsepower",
    y="mpg",
    hue="origin",
    style="origin",
    size="weight",
    palette="Set1"
)
plt.title("Scatter Plot")
plt.tight_layout()
plt.show()
```



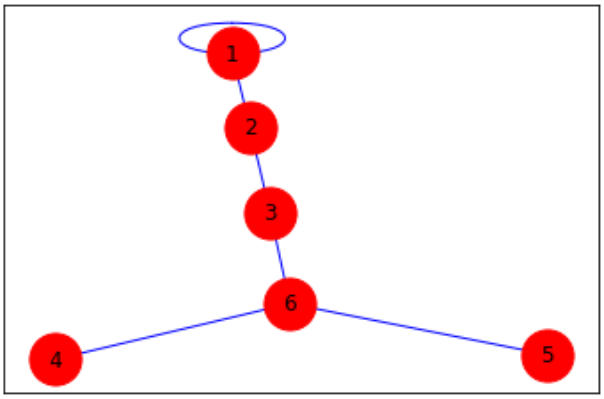
In [35]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sd=pd.read_csv("auto-mpg.csv")
df["horsepower"]=pd.to_numeric(df["horsepower"],errors="coerce")
df=df.dropna(subset=["horsepower"])
plt.figure(figsize=(10,6))

sns.scatterplot(
    data=df,
    x="horsepower",
    y="mpg"
)
plt.title("Scatter Plot")
plt.tight_layout()
plt.show()
```



Visualizing Graphs using NetworkX Library

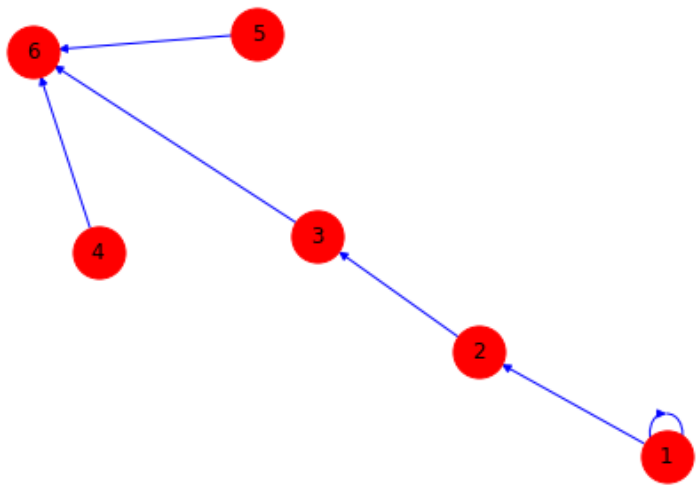
Undirected Graphs

In [6]:
```
import networkx as nx
import matplotlib.pyplot as plt

G=nx.Graph()

G.add_node(1)  # to add one node
G.add_nodes_from([2,3])   # to add multiple node
G.add_nodes_from(range(4,7))

G.add_edge(1,2)
G.add_edge(1,1)
G.add_edges_from([(2,3),(3,6),(4,6),(5,6)])

nx.draw_networkx(G,node_size=850,node_color="red",edge_color="blue",width=1)
#
nx.draw(G,with_labels=True,node_size=850,node_color="red",edge_color="blue",width=1)
plt.show()
```

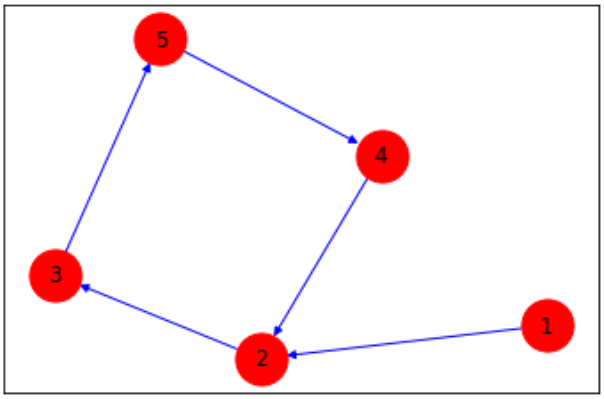


Direction

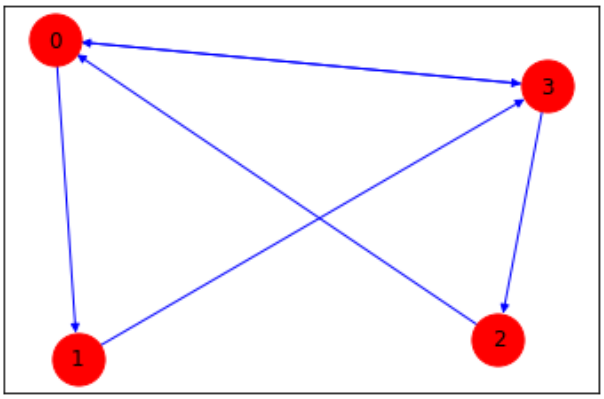
In [19]:
```
import networkx as nx
import matplotlib.pyplot as plt

G=nx.DiGraph()

G.add_node(1)  # to add one node
G.add_nodes_from([2,3])   # to add multiple node
G.add_nodes_from(range(4,7))

G.add_edge(1,2)
G.add_edge(1,1)
G.add_edges_from([(2,3),(3,6),(4,6),(5,6)])

nx.draw(G,with_labels=True,node_size=850,node_color="red",edge_color="blue",width=1)
#  nx.draw_networkx(G,node_size=850,node_color="red",edge_color="blue",width=1)
plt.show()
```



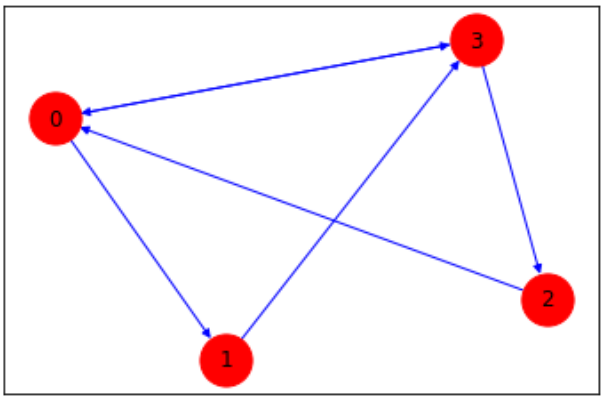
In [18]:
```
import networkx as nx
import matplotlib.pyplot as plt

G=nx.DiGraph()

G.add_nodes_from([1,2,3,4,5])

G.add_edge(1,2)
G.add_edge(4,2)
G.add_edge(3,5)
G.add_edge(2,3)
G.add_edge(5,4)


#nx.draw(G,with_labels=True,node_size=850,node_color="red",edge_color="blue",width=1)
nx.draw_networkx(G,node_size=850,node_color="red",edge_color="blue",width=1)
plt.show()
```



In [17]:
```
import networkx as nx
import matplotlib.pyplot as plt

G=nx.DiGraph()

G.add_nodes_from([0,1,2,3])

G.add_edge(0,3)
G.add_edge(3,0)
G.add_edge(0,1)
G.add_edge(1,3)
G.add_edge(3,2)
G.add_edge(2,0)


#
nx.draw(G,with_labels=True,node_size=850,node_color="red",edge_color="blue",width=1)
nx.draw_networkx(G,node_size=850,node_color="red",edge_color="blue",width=1)
plt.show()
```



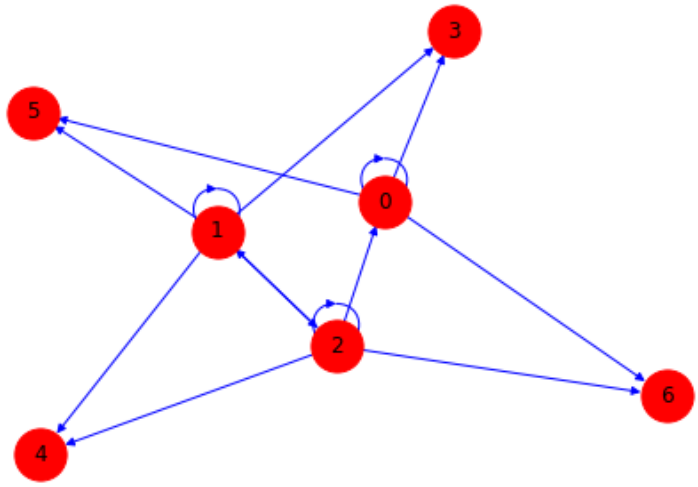
In [20]:
```
import networkx as nx
import matplotlib.pyplot as plt

G=nx.DiGraph()

G.add_nodes_from(range(4))
l=[[0,1,0,1],[0,0,0,1],[1,0,0,0],[1,0,1,0]]

for i in range(4):
    for j in range(4):
        if l[i][j]==1:
            G.add_edge(i,j)

#nx.draw(G,with_labels=True,node_size=850,node_color="red",edge_color="blue",width=1)
nx.draw_networkx(G,node_size=850,node_color="red",edge_color="blue",width=1)
plt.show()
```



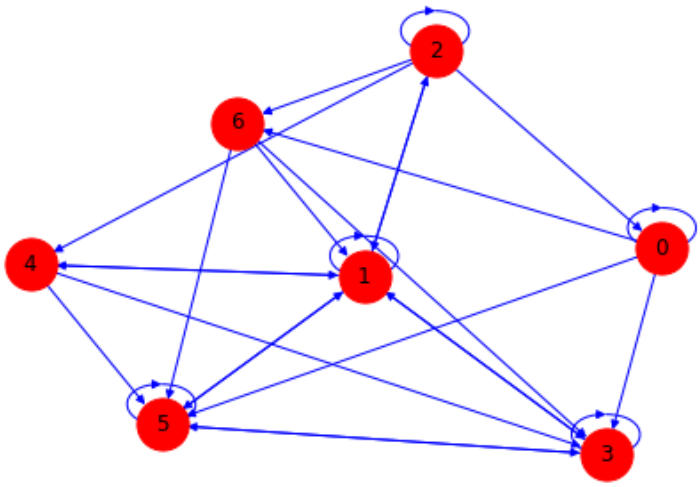
Directed Graph from Adjacency Matrix with 3 nodes

In [36]:

```
import networkx as nx
import matplotlib.pyplot as plt

G=nx.DiGraph()

G.add_nodes_from([0,1,2])
l=[[1,0,0,1,0,1,1],[0,1,1,1,1,1,0],[1,1,1,0,1,0,1]]

for i in range(0,3):
    for j in range(0,7):
        if l[i][j]==1:
            G.add_edge(i,j)

nx.draw(G,with_labels=True,node_size=850,node_color="red",edge_color="blue",width=1)
# nx.draw_networkx(G,node_size=850,node_color="red",edge_color="blue",width=1)
plt.show()
```

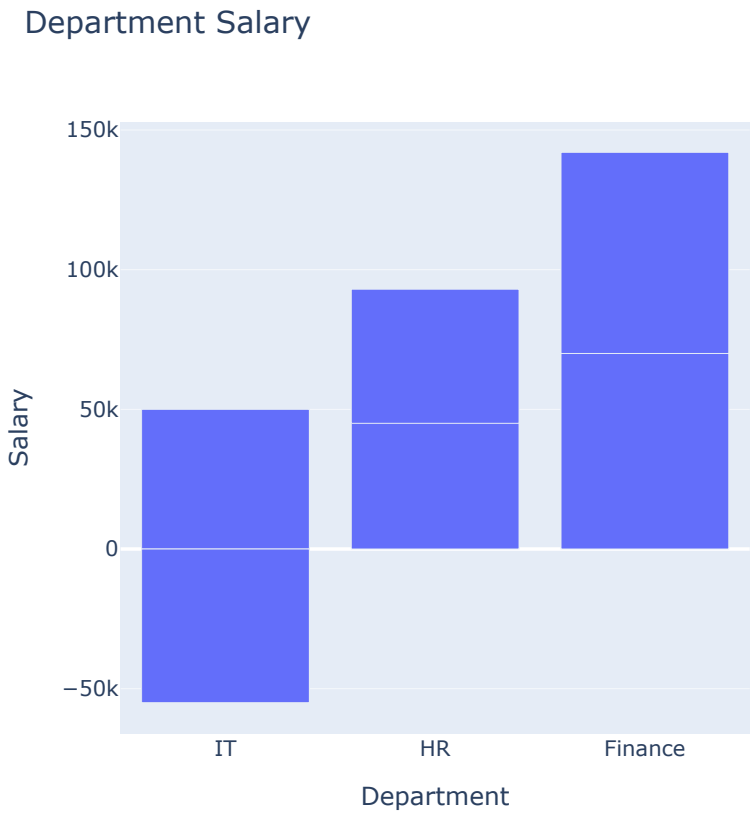


In [38]:
```
import networkx as nx
import matplotlib.pyplot as plt

G=nx.DiGraph()

G.add_nodes_from([0,1,2,3])
l=[[1,0,0,1,0,1,1],[0,1,1,1,1,1,0],[1,1,1,0,1,0,1],[0,1,0,1,0,1,0],[0,1,0,1,0,1,0],
[0,1,0,1,0,1,0],[0,1,0,1,0,1,0]]

for i in range(0,7):
    for j in range(0,7):
        if l[i][j]==1:
            G.add_edge(i,j)

nx.draw(G,with_labels=True,node_size=850,node_color="red",edge_color="blue",width=1)
# nx.draw_networkx(G,node_size=850,node_color="red",edge_color="blue",width=1)
plt.show()
```



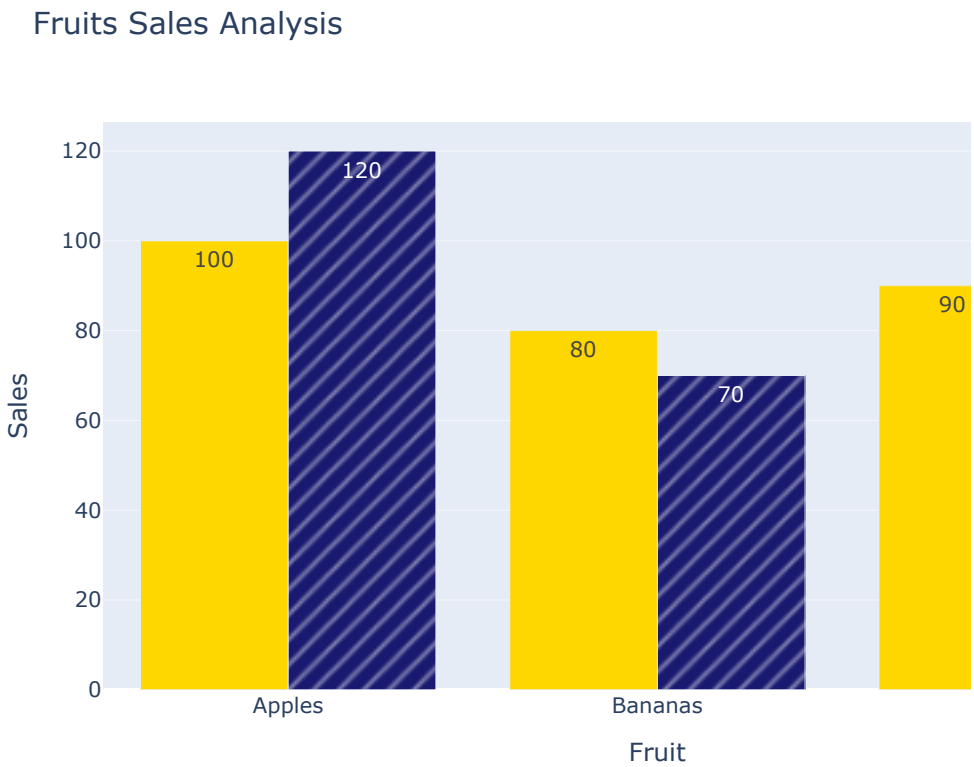
Bar Chart (plotly)

In [42]:
```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default="notebook"

data={"Department":["IT","IT","HR","HR","Finance","Finance"],
     "Gender":["Male","Female","Male","Female","Male","Female"],
     "Salary":[50000,-55000,45000,48000,70000,72000]}

df=pd.DataFrame(data)
fig=px.bar(df,x="Department",y="Salary",title="Department Salary")
fig.show()
```



In [47]:

```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default="notebook"

# sample data
data={
    "Fruit":["Apples","Apples","Bananas","Bananas","Oranges","Oranges"],
    "City":["NYC","LA","NYC","LA","NYC","LA"],
    "Sales":[100,120,80,70,90,110],
    "Profit":[20,25,15,10,18,22],
    "Region":["Eest","West","Eest","West","Eest","West"]
}

fig=px.bar(data,x="Fruit",
           y="Sales",
           width=800,
           height=500,
           title="Fruits Sales Analysis",
           color="City",
           hover_data=["Profit","Region"],
           pattern_shape="City",
           barmode="group",
           text_auto=".2s",
           text="Sales",
           orientation="v",
           color_discrete_map={"NYC":"gold","LA":"midnightblue"}
          )

fig.show()
```

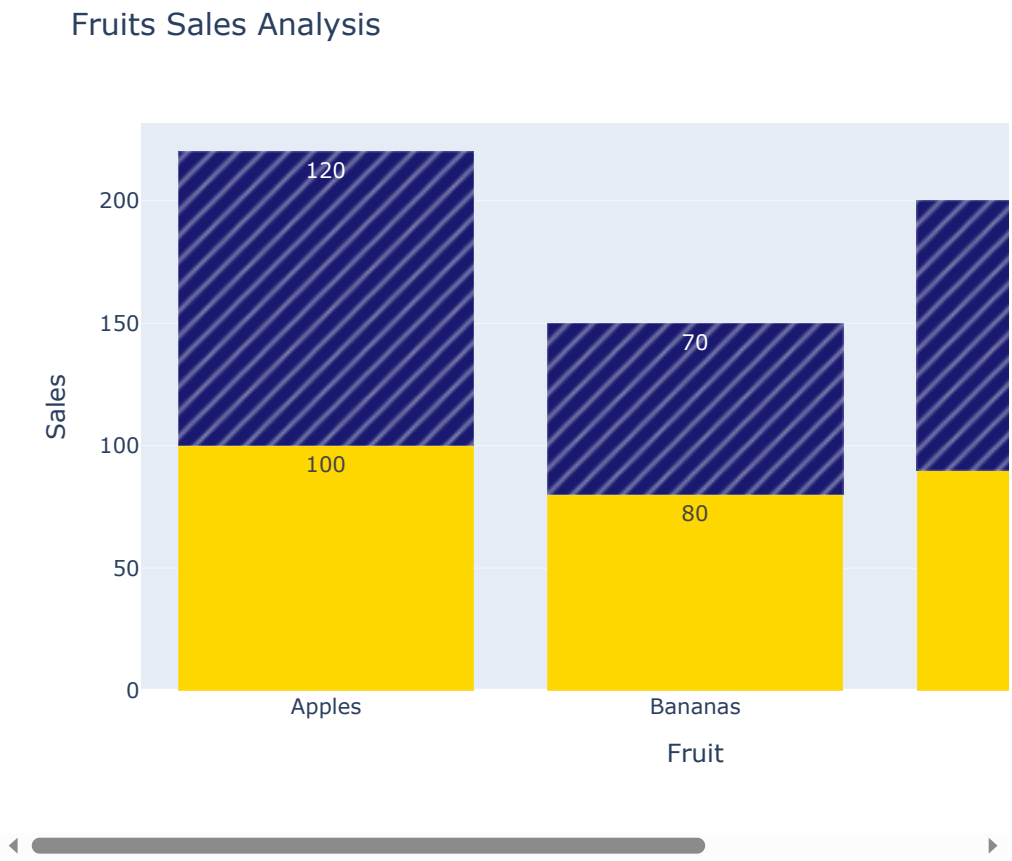


In [50]:

```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default="notebook"

# sample data
data={
    "Fruit":["Apples","Apples","Bananas","Bananas","Oranges","Oranges"],
    "City":["NYC","LA","NYC","LA","NYC","LA"],
    "Sales":[100,120,80,70,90,110],
    "Profit":[20,25,15,10,18,22],
    "Region":["Eest","West","Eest","West","Eest","West"]
}

fig=px.bar(data,x="Fruit",
           y="Sales",
           width=800,
           height=500,
           title="Fruits Sales Analysis",
           color="City",
           hover_data=["Profit","Region"],
           pattern_shape="City",
           barmode="group",
           text_auto=".2s",
           text="Sales",
           orientation="v",
           #color_discrete_map={"NYC":"gold","LA":"midnightblue"}
           color_discrete_sequence=px.colors.qualitative.Pastel #(Optional
alternative)
          )

fig.show()
```

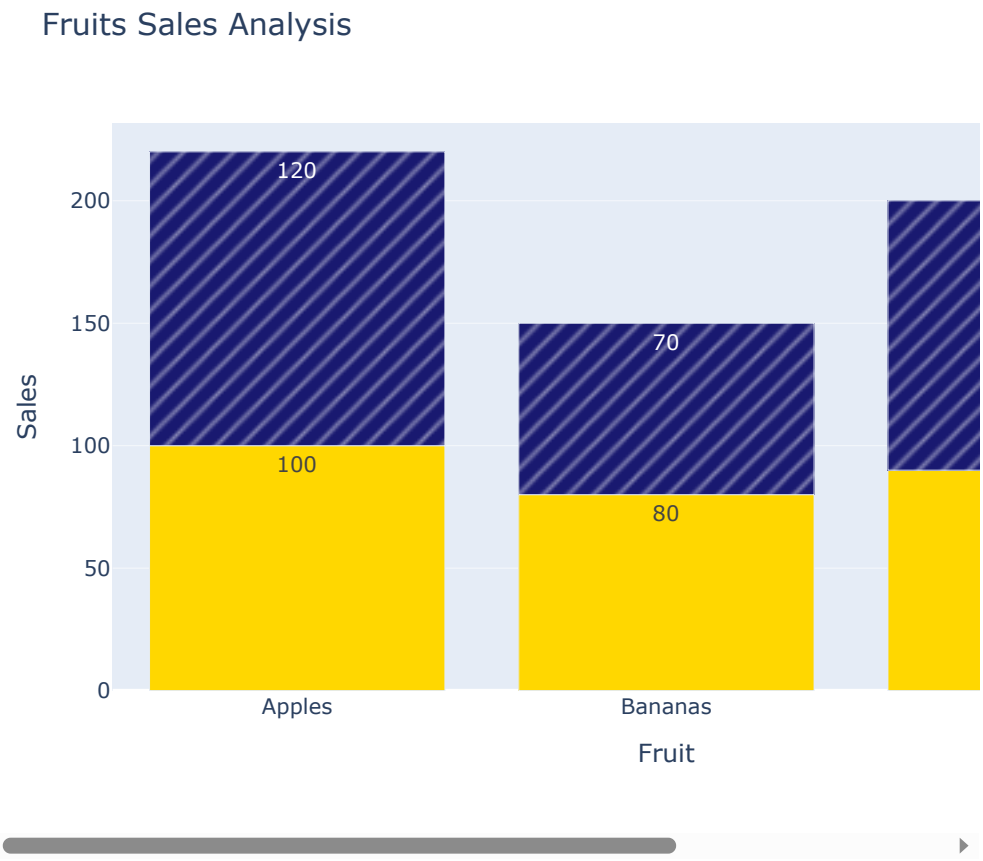
In [52]:

```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default="notebook"

# sample data
data={
    "Fruit":["Apples","Apples","Bananas","Bananas","Oranges","Oranges"],
    "City":["NYC","LA","NYC","LA","NYC","LA"],
    "Sales":[100,120,80,70,90,110],
    "Profit":[20,25,15,10,18,22],
    "Region":["Eest","West","Eest","West","Eest","West"]
}

fig=px.bar(data,x="Fruit",
           y="Sales",
           width=800,
           height=500,
           title="Fruits Sales Analysis",
           color="City",
           hover_data=["Profit","Region"],
           pattern_shape="City",
           barmode="stack",
           text_auto=".2s",
           text="Sales",
           orientation="v",
           color_discrete_map={"NYC":"gold","LA":"midnightblue"}
           #color_discrete_sequence=px.colors.qualitative.Pastel #(Optional
alternative)
          )

fig.show()
```



In [53]:

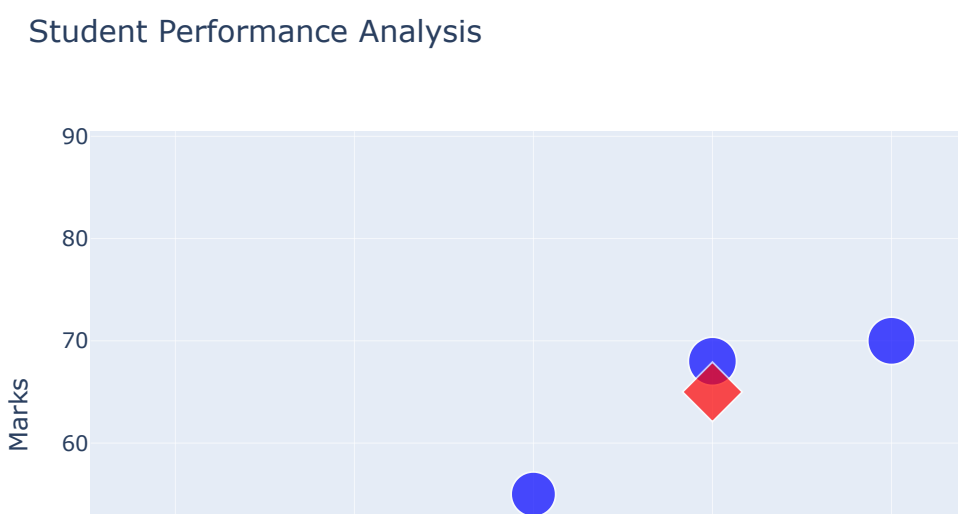
```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default="notebook"

# sample data
data={
    "Fruit":["Apples","Apples","Bananas","Bananas","Oranges","Oranges"],
    "City":["NYC","LA","NYC","LA","NYC","LA"],
    "Sales":[100,120,80,70,90,110],
    "Profit":[20,25,15,10,18,22],
    "Region":["Eest","West","Eest","West","Eest","West"]
}

fig=px.bar(data,x="Fruit",
           y="Sales",
           width=800,
           height=500,
           title="Fruits Sales Analysis",
           color="City",
           hover_data=["Profit","Region"],
           pattern_shape="City",
           barmode="relative",
           text_auto=".2s",
           text="Sales",
           orientation="v",
           color_discrete_map={"NYC":"gold","LA":"midnightblue"}
           #color_discrete_sequence=px.colors.qualitative.Pastel #(Optional
alternative)
          )

fig.show()
```

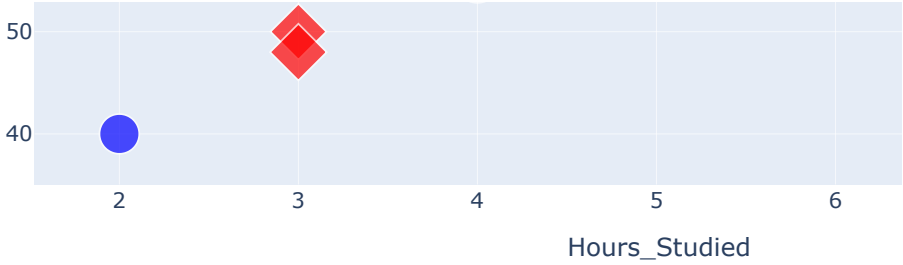


Scatter (Plotly)

In [54]:

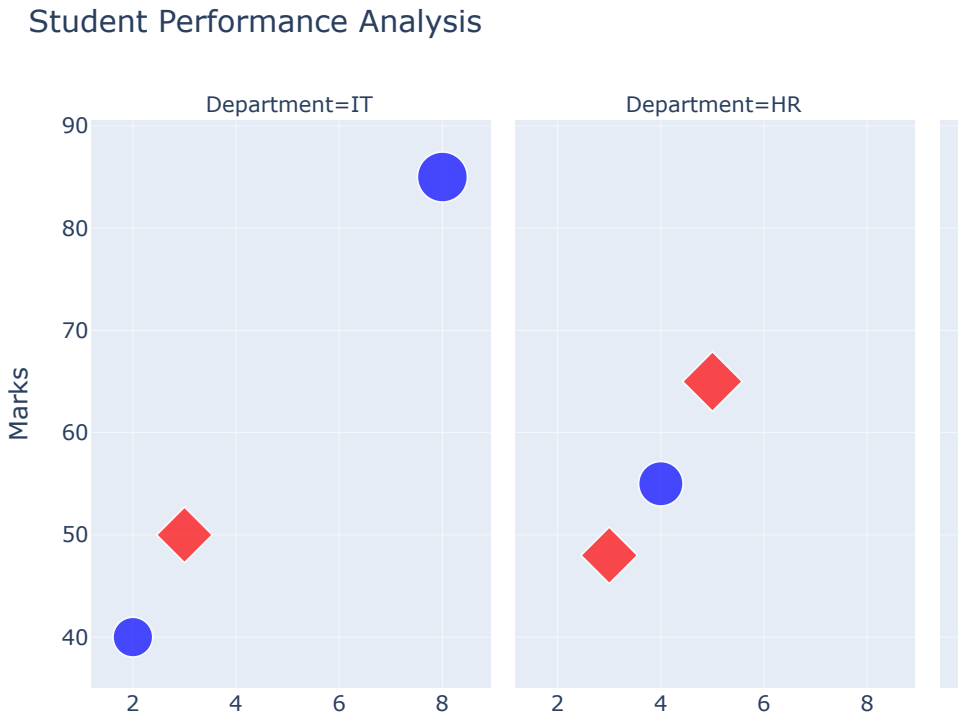
```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default='notebook'

data = {
    "Hours_Studied": [2,3,4,5,6,7,8,3,5],
    "Marks": [40,50,55,65,70,80,85,48,68],
    "Gender": ["Male","Female","Male","Female","Male","Female","Male",
               "Female","Male"],
    "Attendance": [60,70,75,80,85,90,95,72,88],
    "Department": ["IT","IT","HR","HR","Finance","Finance","IT","HR","Finance"]
}

df = pd.DataFrame(data)
fig = px.scatter(
    df,
    x="Hours_Studied",
    y="Marks",
    width=900,
    height=500,
    title="Student Performance Analysis",
    color="Gender",                 # grouping by color
    size="Attendance",              # bubble size
    color_discrete_map={
        "Male": "blue",
        "Female": "red"
    },

#     color_discrete_sequence=["blue","red"],

    symbol="Gender",                # different marker shapes

#     facet_col="Department",         # separate charts by column
#     facet_row="Gender"              # separate charts by row
)

fig.show()
```




50
40
2
3
4
5
6
Hours_Studied

In [55]:

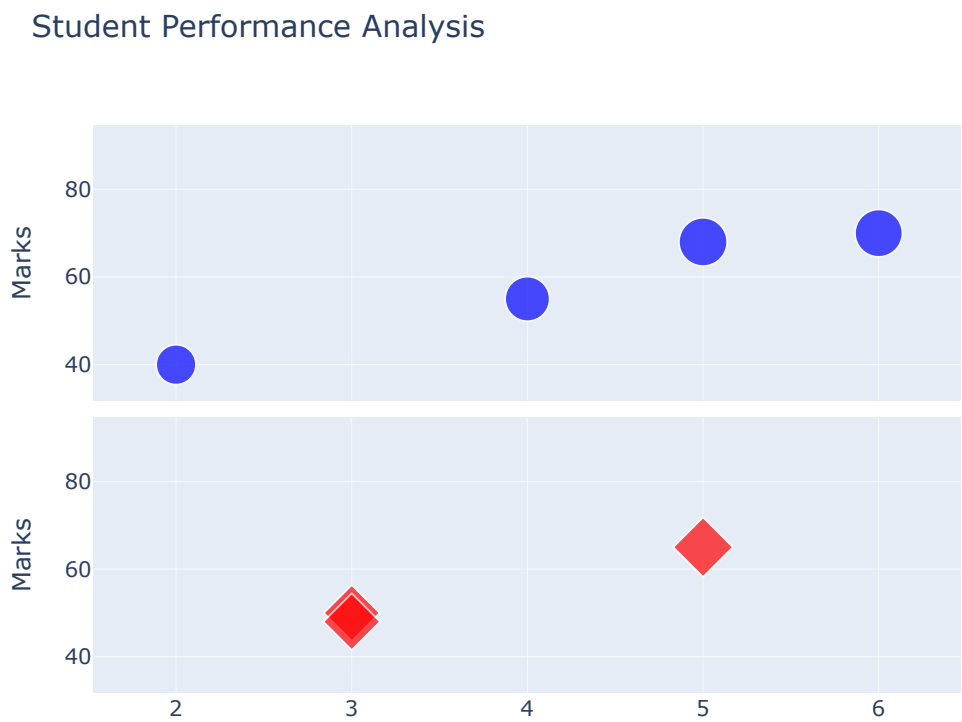
```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default='notebook'

data = {
    "Hours_Studied": [2,3,4,5,6,7,8,3,5],
    "Marks": [40,50,55,65,70,80,85,48,68],
    "Gender": ["Male","Female","Male","Female","Male","Female","Male",
               "Female","Male"],
    "Attendance": [60,70,75,80,85,90,95,72,88],
    "Department": ["IT","IT","HR","HR","Finance","Finance","IT","HR","Finance"]
}

df = pd.DataFrame(data)
fig = px.scatter(
    df,
    x="Hours_Studied",
    y="Marks",
    width=900,
    height=500,
    title="Student Performance Analysis",
    color="Gender",                 # grouping by color
    size="Attendance",              # bubble size
    color_discrete_map={
        "Male": "blue",
        "Female": "red"
    },

#     color_discrete_sequence=["blue","red"],

    symbol="Gender",                # different marker shapes

     facet_col="Department",         # separate charts by column
#     facet_row="Gender"              # separate charts by row
)

fig.show()
```



Hours_Studied Hours_Studied

In [56]:

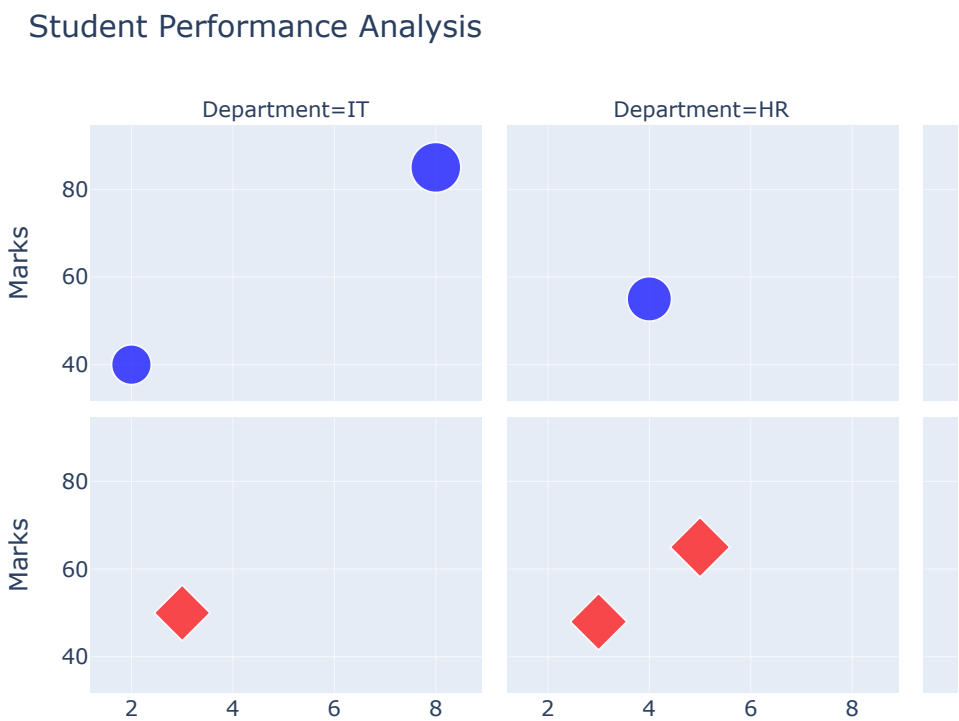
```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default='notebook'

data = {
    "Hours_Studied": [2,3,4,5,6,7,8,3,5],
    "Marks": [40,50,55,65,70,80,85,48,68],
    "Gender": ["Male","Female","Male","Female","Male","Female","Male",
               "Female","Male"],
    "Attendance": [60,70,75,80,85,90,95,72,88],
    "Department": ["IT","IT","HR","HR","Finance","Finance","IT","HR","Finance"]
}

df = pd.DataFrame(data)
fig = px.scatter(
    df,
    x="Hours_Studied",
    y="Marks",
    width=900,
    height=500,
    title="Student Performance Analysis",
    color="Gender",                 # grouping by color
    size="Attendance",              # bubble size
    color_discrete_map={
        "Male": "blue",
        "Female": "red"
    },

#     color_discrete_sequence=["blue","red"],

    symbol="Gender",                # different marker shapes

#     facet_col="Department",         # separate charts by column
     facet_row="Gender"              # separate charts by row
)

fig.show()
```



Hours_Studied

In [57]:

```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default='notebook'

data = {
    "Hours_Studied": [2,3,4,5,6,7,8,3,5],
    "Marks": [40,50,55,65,70,80,85,48,68],
    "Gender": ["Male","Female","Male","Female","Male","Female","Male",
               "Female","Male"],
    "Attendance": [60,70,75,80,85,90,95,72,88],
    "Department": ["IT","IT","HR","HR","Finance","Finance","IT","HR","Finance"]
}

df = pd.DataFrame(data)
fig = px.scatter(
    df,
    x="Hours_Studied",
    y="Marks",
    width=900,
    height=500,
    title="Student Performance Analysis",
    color="Gender",                 # grouping by color
    size="Attendance",              # bubble size
    color_discrete_map={
        "Male": "blue",
        "Female": "red"
    },

#     color_discrete_sequence=["blue","red"],

    symbol="Gender",                # different marker shapes

     facet_col="Department",         # separate charts by column
     facet_row="Gender"              # separate charts by row
)

fig.show()
```



Hours_Studied Hours_Studied

In [58]:

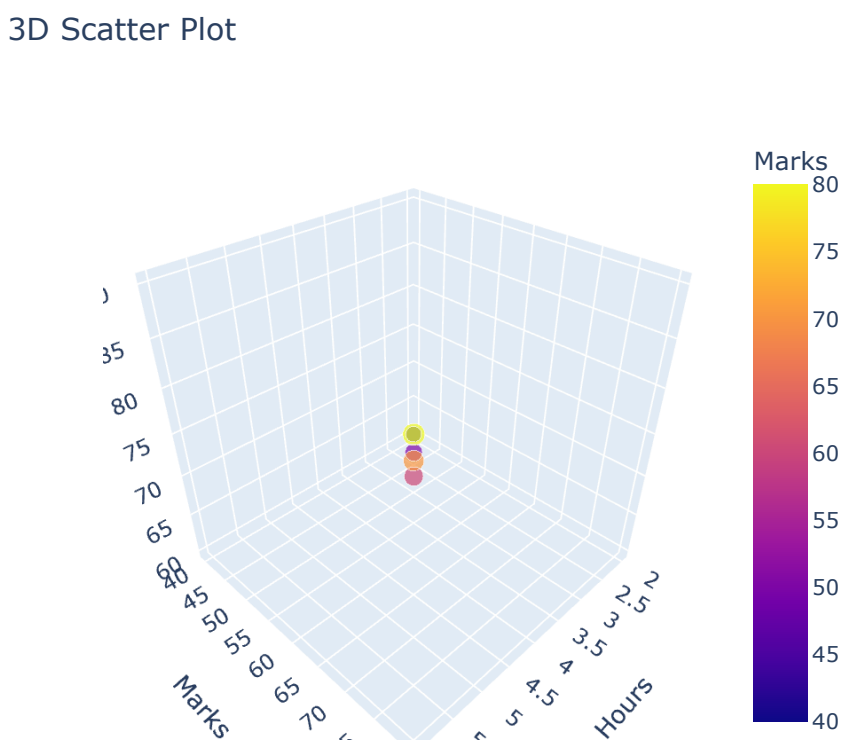
```
import pandas as pd
import plotly.express as px
import plotly.io as pio
pio.renderers.default='notebook'

data = {
    "Hours_Studied": [2,3,4,5,6,7,8,3,5],
    "Marks": [40,50,55,65,70,80,85,48,68],
    "Gender": ["Male","Female","Male","Female","Male","Female","Male",
               "Female","Male"],
    "Attendance": [60,70,75,80,85,90,95,72,88],
    "Department": ["IT","IT","HR","HR","Finance","Finance","IT","HR","Finance"]
}

df = pd.DataFrame(data)
fig = px.scatter(
    df,
    x="Hours_Studied",
    y="Marks",
    width=900,
    height=500,
    title="Student Performance Analysis",
    color="Gender",                 # grouping by color
    size="Attendance",              # bubble size
    color_discrete_map={
        "Male": "blue",
        "Female": "red"
    },
         color_discrete_sequence=["blue","red"],

    symbol="Gender",                # different marker shapes

#     facet_col="Department",         # separate charts by column
#     facet_row="Gender"              # separate charts by row
)

fig.show()
```

Hours_Studied

3-D plot

In [61]:

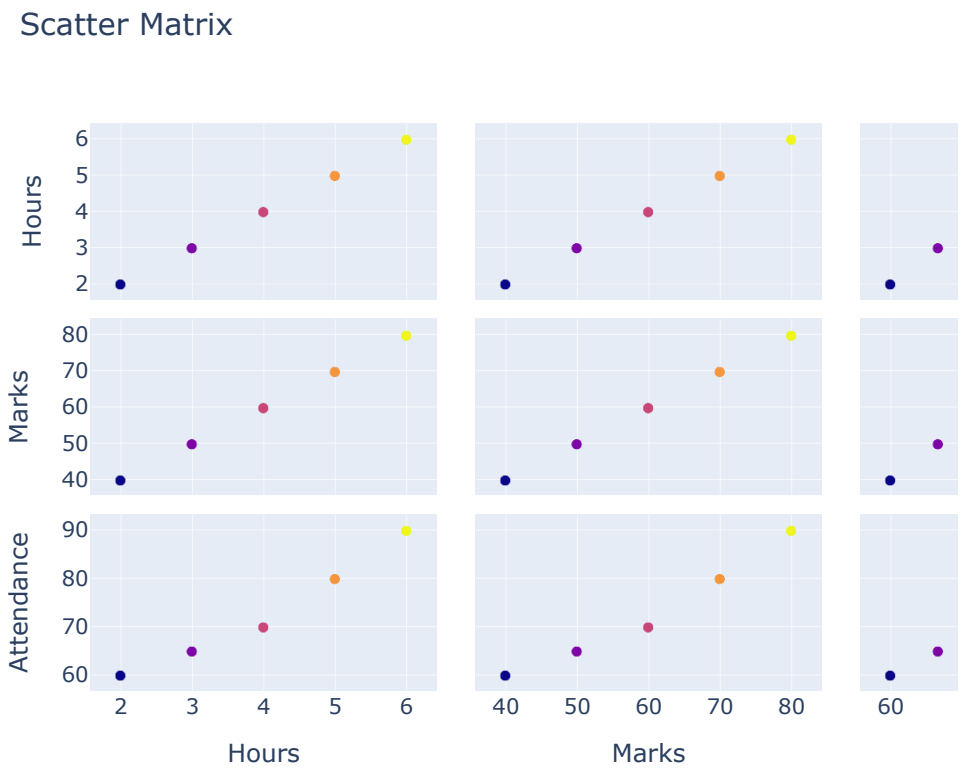
```
import pandas as pd
import plotly.express as px

import plotly.io as pio
pio.renderers.default='notebook'

data = {
    "Hours": [2,3,4,5,6],
    "Marks": [40,50,60,70,80],
    "Attendance": [60,65,70,80,90]
}
df = pd.DataFrame(data)

fig = px.scatter_3d(
    df,
    x="Hours",
    y="Marks",
    z="Attendance",
    color="Marks",
    title="3D Scatter Plot",
    size='Marks'
)

fig.show()
```



Scatter matrix

In [66]:

```
import pandas as pd
import plotly.express as px
import plotly.io as pio

pio.renderers.default='notebook'

data = {
    "Hours": [2,3,4,5,6],
    "Marks": [40,50,60,70,80],
    "Attendance": [60,65,70,80,90]
}
df = pd.DataFrame(data)
fig = px.scatter_matrix(
    df,
    dimensions=["Hours","Marks","Attendance"],
    color="Marks",
    title="Scatter Matrix",
    width=800,
    height=500
)

fig.show()
```



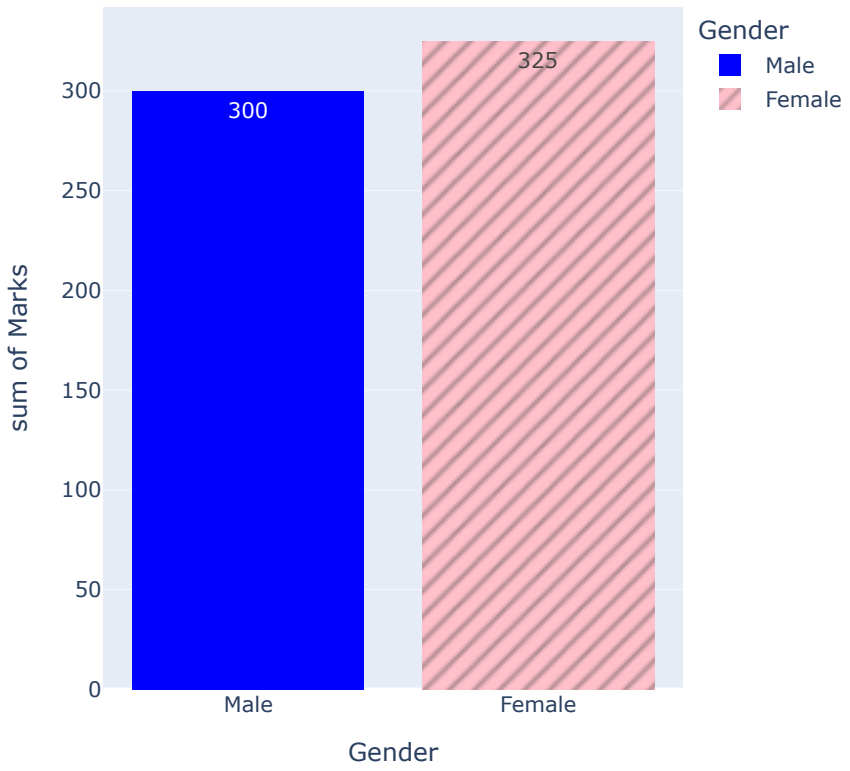
Line plot

In [67]:

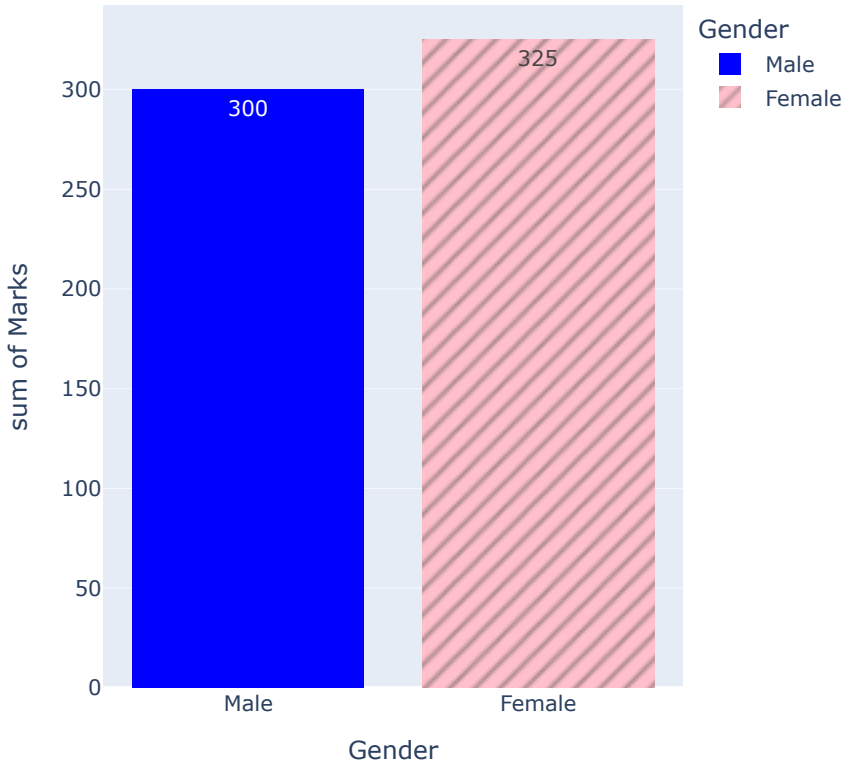
```
import pandas as pd
import plotly.express as px

data = {
    "Month": ["Jan","Feb","Mar","Apr","Jan","Feb","Mar","Apr"],
    "Sales": [100,120,140,160,90,110,130,150],
    "Region": ["North","North","North","North",
               "South","South","South","South"]
}

df = pd.DataFrame(data)
fig = px.line(
    df,
    x="Month",
    y="Sales",
    color="Region",
    markers=True,
    width=800,
    height=500,
    title="Monthly Sales Comparison",
    color_discrete_sequence=["blue","red"]
)

fig.show()
```

Histogram

In [68]:
```
data = {
    "Marks":[40,45,50,55,60,65,70,75,80,85],
    "Gender":["Male","Female","Male","Female","Male",
              "Female","Male","Female","Male","Female"]
}

df = pd.DataFrame(data)
fig = px.histogram(
    df,
    x="Gender",
    y='Marks',
    nbins=5,
    color="Gender",
    histfunc="sum",
    pattern_shape="Gender",
    text_auto=True,
    color_discrete_map={"Male":"blue","Female":"pink"},
#     color_discrete_sequence=["blue","pink"] #Optional
)

fig.show()
```

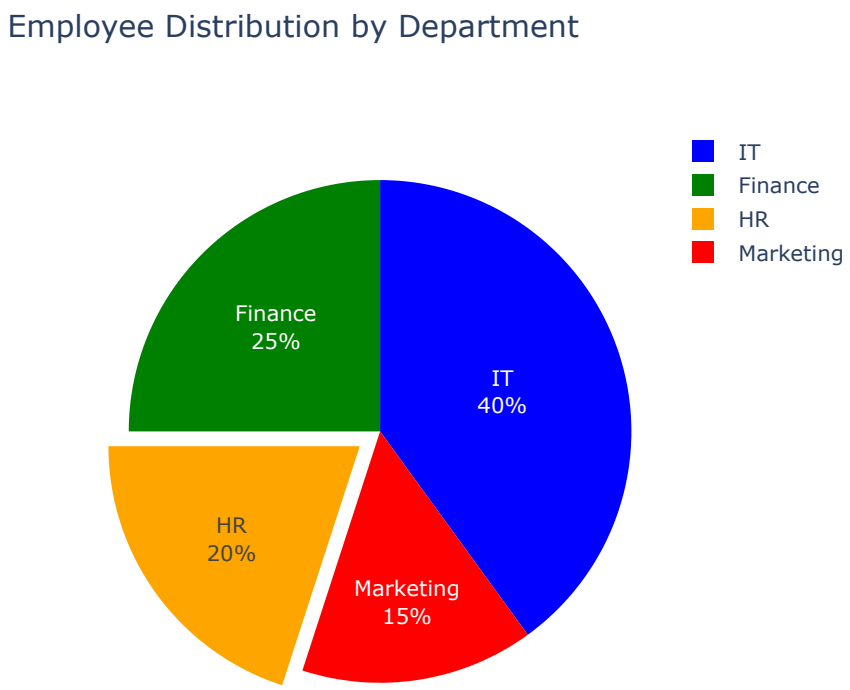


In [69]:
```
data = {
    "Marks":[40,45,50,55,60,65,70,75,80,85],
    "Gender":["Male","Female","Male","Female","Male",
              "Female","Male","Female","Male","Female"]
}

df = pd.DataFrame(data)
fig = px.histogram(
    df,
    x="Gender",
    y='Marks',
    nbins=5,
    color="Gender",
    histfunc="sum",
    pattern_shape="Gender",
    text_auto=True,
#    color_discrete_map={"Male":"blue","Female":"pink"},
    color_discrete_sequence=["blue","pink"] #Optional
)

fig.show()
```



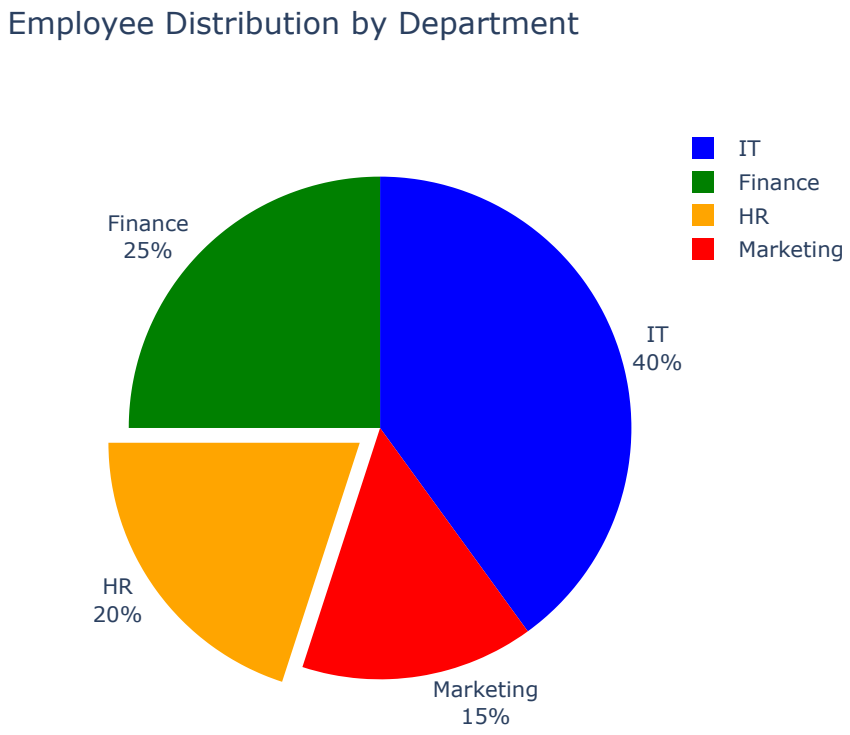
Pie Chart (Plotly)

In [71]:
```
import pandas as pd
import plotly.express as px

data = {
    "Department": ["IT", "HR", "Finance", "Marketing"],
    "Employees": [40, 20, 25, 15]
}

df = pd.DataFrame(data)
fig = px.pie(
    df,
    values="Employees",
    names="Department",
    color_discrete_sequence=["blue", "green", "orange", "red"],
    title="Employee Distribution by Department"
)

fig.update_traces(
    textinfo="percent+label",
    textposition="inside",
    pull=[0,0.1,0,0]
)
fig.show()
```



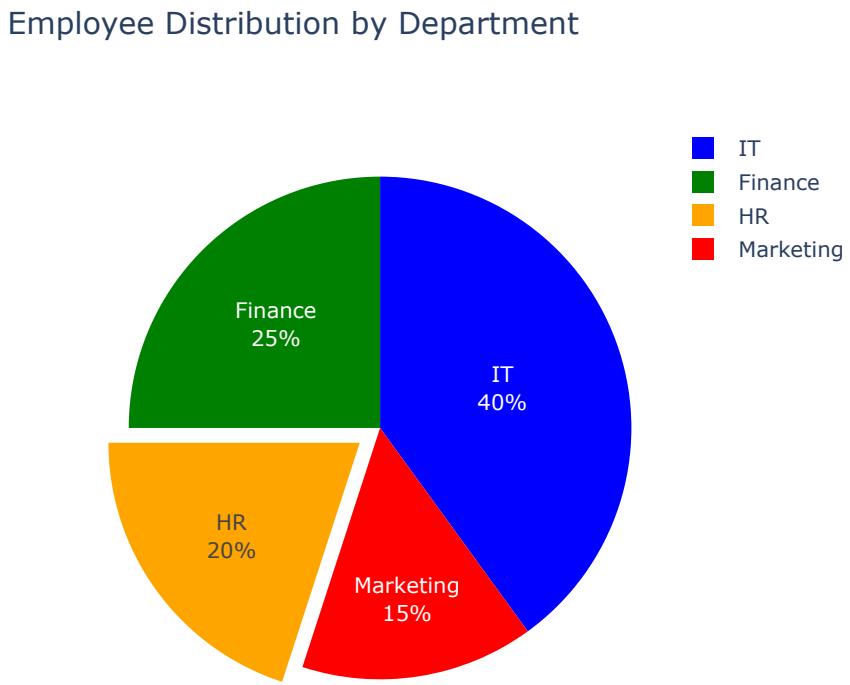
In [72]:

```
import pandas as pd
import plotly.express as px

data = {
    "Department": ["IT", "HR", "Finance", "Marketing"],
    "Employees": [40, 20, 25, 15]
}

df = pd.DataFrame(data)
fig = px.pie(
    df,
    values="Employees",
    names="Department",
    color_discrete_sequence=["blue", "green", "orange", "red"],
    title="Employee Distribution by Department"
)

fig.update_traces(
    textinfo="percent+label",
    textposition="outside",
    pull=[0,0.1,0,0]
)
fig.show()
```



In [73]:

```
import pandas as pd
import plotly.express as px

data = {
    "Department": ["IT", "HR", "Finance", "Marketing"],
    "Employees": [40, 20, 25, 15]
}

df = pd.DataFrame(data)
fig = px.pie(
    df,
    values="Employees",
    names="Department",
    color_discrete_sequence=["blue", "green", "orange", "red"],
    title="Employee Distribution by Department"
)

fig.update_traces(
    textinfo="percent+label",
    textposition="auto",
    pull=[0,0.1,0,0]
)
fig.show()
```



In [ ]: